# OverTheWire – Writeup

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**Bandit Level 0**
https://overthewire.org/wargames/bandit/bandit0.html
**Task:**
The goal of this level is for we to log into the game using SSH. The host to which we need to connect is **bandit.labs.overthewire.org**, on port 2220. The username is **bandit0** and the password is **bandit0**.
**Solution:**
Log in using this command
ssh bandit0@bandit.labs.overthewire.org -p 2220
then enter password: bandit0 (password is hidden always)

- ssh - Secure Shell, for remote log in to another system.
- bandit0 - Username for the Bandit level 0 account.
- bandit.labs.overthewire.org - Remote server address.
- -p 2220 - Specifies port number (Bandit uses port 2220 instead of the default SSH port 22).

**Username:** bandit0
**Password:** bandit0

**Bandit Level 0 - Level 1**
**Task:**
Find password that is located in a file called readme to proceed to next levels
**Solution:**

```
bandit0@bandit:~$ ls
readme
bandit0@bandit:~$ cat readme
Congratulations on your first steps into the bandit game!!
Please make sure you have read the rules at https://overthewire.org/rules/
If you are following a course, workshop, walkthrough or other educational activity,
please inform the instructor about the rules as well and encourage them to
contribute to the OverTheWire community so we can keep these games free!

The password you are looking for is: ZjLjTmM6FvvyRnrb2rfNWOZOTa6ip5If

bandit0@bandit:~$
```

ls - Shows available files
cat readme - Reads the file content
**Username:** bandit1
**Password:** ZjLjTmM6FvvyRnrb2rfNWOZOTa6ip5If

**Bandit Level 1 - Level 2**
**Task:**
The password for the next level is stored in a file called **-** located in the home directory
**Solution:**
Log in using : ssh bandit1@bandit.labs.overthewire.org -p 2220

```
bandit1@bandit:~$ ls
-
bandit1@bandit:~$ cat <-
263JGJPfgU6LtdEvgfWU1XP5yac29mFx
bandit1@bandit:~$
```

- is treated as **standard input**, not a normal filename.
cat <- tells cat to read the file named -.
OR cat ./- (Special characters confuse Linux, ./ tells Linux: "this is a file here")

**Why:**
The filename is -, so it must be referenced explicitly to read its content.
**Username:** bandit2
**Password:** 263JGJPfgU6LtdEvgfWU1XP5yac29mFx

**Bandit Level 2 - Level 3**
**Task:**
The password for the next level is stored in a file called --spaces in this filename-- located in the home directory.
**Solution:**
Log in using : ssh bandit2@bandit.labs.overthewire.org -p 2220

```
bandit2@bandit:~$ ls
--spaces in this filename--
bandit2@bandit:~$ cat "./--spaces in this filename--"
MNk8KNH3Usiio41PRUEoDFPqfxLPlSmx
bandit2@bandit:~$
```

- ls - Used to see the exact filename, including spaces.
- cat "./--spaces in this filename--" - Quotes are needed because spaces break filenames.

**Why:**
Quoting tells the shell to treat the whole name as one file.
**Username:** bandit3
**Password:** MNk8KNH3Usiio41PRUEoDFPqfxLPlSmx

**Bandit Level 3 - Level 4**
**Task:**
The password for the next level is stored in a hidden file in the **inhere** directory.
**Solution:**
Log in using : ssh bandit3@bandit.labs.overthewire.org -p 2220

```
bandit3@bandit:~$ ls -a
.  ..  .bash_logout  .bashrc  inhere  .profile
bandit3@bandit:~$ ls
inhere
bandit3@bandit:~$ cd inhere
bandit3@bandit:~/inhere$ ls -a
.  ..  ...Hiding-From-You
bandit3@bandit:~/inhere$ cat ...Hiding-From-You
2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ
bandit3@bandit:~/inhere$
```

**All hidden files and folders in linux are stored with a dot in front of their name.**

- ls -a - Shows hidden files (starting with .)
- Normal ls - Hides those files
- cd inhere - Moves into the directory
- cat ...Hiding-From-We - Reads the hidden file

**Why:**
The password is hidden, so ls -a is required to find it.
**Username:** bandit4
**Password:** 2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ

**Bandit Level 4 - Level 5**
**Task:**
The password for the next level is stored in the only human-readable file in the **inhere** directory. Tip: if your terminal is messed up, try the “reset” command.
**Solution:**
Log in using : ssh bandit4@bandit.labs.overthewire.org -p 2220
IMPORTANT: Files starting with -
Linux thinks they are options, not filenames.
So we must use ./
**WHY:**
file - tells file type
/* - all files in current directory
The file marked **ASCII text** is **human-readable**.

```
bandit4@bandit:~$ ls
inhere
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ ls
-file00  -file01  -file02  -file03  -file04  -file05  -file06  -file07  -file08  -file09
bandit4@bandit:~/inhere$ file ./*
./-file00: data
./-file01: OpenPGP Public Key
./-file02: OpenPGP Public Key
./-file03: data
./-file04: data
./-file05: data
./-file06: data
./-file07: ASCII text
./-file08: data
./-file09: data
bandit4@bandit:~/inhere$ cat ./-file07
4oQYVPkxZOOEOO5pTW81FB8j8lxXGUQw
```

**Username:** bandit5
**Password:** 4oQYVPkxZOOEOO5pTW81FB8j8lxXGUQw

**Bandit Level 5 - Level 6**
**Task:**
The password for the next level is stored in a file somewhere under the **inhere** directory and has all of the following properties: **human-readable, 1033 bytes in size, not executable**
**Solution:**
Log in using : ssh bandit5@bandit.labs.overthewire.org -p 2220
**Find:** Instead of checking one by one, we search smartly. This command filters only one file that matches all rules.

```
bandit5@bandit:~$ ls
inhere
bandit5@bandit:~$ cd inhere
bandit5@bandit:~/inhere$ ls
maybehere00  maybehere03  maybehere06  maybehere09  maybehere12  maybehere15  maybehere18
maybehere01  maybehere04  maybehere07  maybehere10  maybehere13  maybehere16  maybehere19
maybehere02  maybehere05  maybehere08  maybehere11  maybehere14  maybehere17
bandit5@bandit:~/inhere$ find . -type f -size 1033c ! -executable
./maybehere07/.file2
bandit5@bandit:~/inhere$ cat ./maybehere07/.file2
HWasnPhtq9AVKe0dmk45nxy20cvUa6EG
```

find . searches from the current directory
-type f selects only regular files
-size 1033c filters files of exactly 1033 bytes
! -executable excludes executable files

**Why:**
To find the only file with these exact properties, which contains the password.
**Username:** bandit6
**Password:** HWasnPhtq9AVKe0dmk45nxy20cvUa6EG

**Bandit Level 6 - Level 7**
**Task:**
The password for the next level is stored **somewhere on the server** and has all of the following properties: owned by user bandit7, owned by group bandit6 and 33 bytes in size
**Solution:**
Log in using : ssh bandit6@bandit.labs.overthewire.org -p 2220

```
bandit6@bandit:~$ find / -type f -user bandit7 -group bandit6 -size 33c 2>/dev/null
/var/lib/dpkg/info/bandit7.password
bandit6@bandit:~$ cat /var/lib/dpkg/info/bandit7.password
morbNTDkSW6jIlUc0ymOdMaLnOlFVAaj
```

| Part | Meaning |
|---|---|
| find / | Search from root (entire server) |
| -type f | Only files |
| -user bandit7 | Owned by user bandit7 |
| -group bandit6 | Group owner bandit6 |
| -size 33c | Exactly 33 bytes |
| 2>/dev/null | Hide permission errors |

**Username:** bandit7
**Password:** morbNTDkSW6jIlUc0ymOdMaLnOlFVAaj

**Bandit Level 7 - Level 8**
**Task:**
The password for the next level is stored in the file **data.txt** next to the word **millionth**
**Solution:**
Log in using : ssh bandit7@bandit.labs.overthewire.org -p 2220
**grep = search text inside files**

```
bandit7@bandit:~$ ls
data.txt
bandit7@bandit:~$ grep millionth data.txt
millionth       dfwvzFQi4mU0wfNbFOe9RoWskMLg7eEc
```

The **grep** command searches for the keyword millionth inside the file **data.txt** and displays the matching line.
**Username:** bandit8
**Password:** dfwvzFQi4mU0wfNbFOe9RoWskMLg7eEc

**Bandit Level 8 - Level 9**
**Task:**
The password for the next level is stored in the file **data.txt** and is the only line of text that occurs only once
**Solution:**
Log in using : ssh bandit8@bandit.labs.overthewire.org -p 2220
**A pipe means:**
"Take output of one command and send it as input to the next command". This is core Linux poyour.

```
bandit8@bandit:~$ ls
data.txt
bandit8@bandit:~$ sort data.txt | uniq -u
4CKMh1JI91bUIZZPXDqGanal4xvAg0JM
bandit8@bandit:~$
```

**uniq works only on sorted data. uniq =** filter duplicate lines….. **-u =** show unique lines only
**sort arranges identical lines together**
**| (pipe):** Sends output of sort - into uniq
**Username:** bandit9
**Password:** 4CKMh1JI91bUIZZPXDqGanal4xvAg0JM

**Bandit Level 9 - Level 10**
**Task:**
The password for the next level is stored in the file **data.txt** in one of the few human-readable strings, preceded by several '=' characters.
**Solution:**
Log in using : ssh bandit9@bandit.labs.overthewire.org -p 2220

```
bandit9@bandit:~$ strings data.txt | grep "="
========== the
9=H`
[!#=Z
========== password
xWf=
f\Z'========== is
e=i{\#
/1=s
nS=F
M=Sl
=LGT
y =1
========== FGUW5ilLVJrxX9kMYMmlN4MgbpfMiqey
'+Y=+
```

- strings data.txt removes binary junk and keeps only readable text.
- | (pipe) sends that output to the next command.
- grep "=" filters and shows only lines that contain =.

**Username:** bandit10
**Password:** FGUW5ilLVJrxX9kMYMmlN4MgbpfMiqey

**Bandit Level 10 - Level 11**
**Task:**
The password for the next level is stored in the file **data.txt**, which contains base64 encoded data
**Solution:**
Log in using : ssh bandit10@bandit.labs.overthewire.org -p 2220
Base64 encoding is a way of converting binary data into readable text.
To get the original password, we must decode the Base64 content.

```
bandit10@bandit:~$ ls
data.txt
bandit10@bandit:~$ cat data.txt
VGhlIHBhc3N3b3JkIGlzIGR0UjE3M2ZaS2IwUlJzREZTR3NnMlJXbnBOVmozcVJyCg==
bandit10@bandit:~$ base64 -d data.txt
The password is dtR173fZKb0RRsDFSGsg2RWnpNVj3qRr
```

base64: This is the Linux command used to encode or decode Base64 data.
-d means decode.

- Without -d - it encodes text into Base64
- With -d - it decodes Base64 back to original text

Since data.txt already contains Base64 encoded data, we must decode it.
**Username:** bandit11
**Password:** dtR173fZKb0RRsDFSGsg2RWnpNVj3qRr

**Bandit Level 11 - Level 12**
**Task:**
The password for the next level is stored in the file **data.txt**, where all loyourcase (a-z) and uppercase (A-Z) letters have been rotated by 13 positions
**Solution:**
Log in using : ssh bandit11@bandit.labs.overthewire.org -p 2220
**ROT13 shifts each alphabet character 13 positions forward. ROT13 is reversible — applying it again gives the original text.**

```
bandit11@bandit:~$ ls
data.txt
bandit11@bandit:~$ cat data.txt
Gur cnffjbeq vf 7k16JArUVv5LxVuJfsSVdbbtaHGlw9D4
bandit11@bandit:~$ tr 'A-Za-z' 'N-ZA-Mn-za-m' < data.txt
The password is 7x16WNeHIi5YkIhWsfFIqoognUTyj9Q4
```

**The tr command is used to apply ROT13 by shifting all alphabetic characters 13 positions, revealing the original password.**
A-Z - N-ZA-M (uppercase rotation)
a-z - n-za-m (loyourcase rotation)
tr replaces each character accordingly
**Username:** bandit12
**Password:** 7x16WNeHIi5YkIhWsfFIqoognUTyj9Q4

**Bandit Level 12 - Level 13**
**Task:**
The password for the next level is stored in data.txt, which is:

- A hexdump of a file
- The file has been compressed multiple times using different formats

We must reverse everything step by step to get the password.
**Concept Used**
Hexdump - Binary (xxd)
Identify file type (file)
Decompress repeatedly (gzip, bzip2, tar)
Use a temporary working directory (mktemp -d)
**Solution:**
Log in using : ssh bandit12@bandit.labs.overthewire.org -p 2220

```
bandit12@bandit:~$ cd /tmp
bandit12@bandit:/tmp$ mktemp -d
/tmp/tmp.vu4qlQbDno
bandit12@bandit:/tmp$ cd /tmp/tmp.vu4qlQbDno
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ cp ~/data.txt .
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ xxd -r data.txt data.bin
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ file data.bin
data.bin: gzip compressed data, was "data2.bin", last modified: Tue Oct 14 09:26:06 2025, max compression, from Uni
x, original size modulo 2^32 564
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ mv data.bin data.gz
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ gunzip data.gz
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ file data
data: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ mv data data.bz2
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ bunzip2 data.bz2
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ file data
data: gzip compressed data, was "data4.bin", last modified: Tue Oct 14 09:26:06 2025, max compression, from Unix, o
riginal size modulo 2^32 20480
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ mv data data.gz
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ gunzip data.gz
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ file data
data: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ mv data data.tar
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ tar -xf data.tar
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ file data
data: cannot open `data' (No such file or directory)
bandit12@bandit:/tmp/tmp.vu4qlQbDno$ ls
data5.bin  data.tar  data.txt
```

**Extract based on type**

- **gzip - mv data data.gz && gunzip data.gz**
- **bzip2 - mv data data.bz2 && bunzip2 data.bz2**
- **tar - mv data data.tar && tar -xf data.tar**

**Username:** bandit13
**Password:** FO5dwFsc0cbaIiH0h8J2eUks2vdTDwAn

**Bandit Level 13 - Level 14**
**Task:**
The password for the next level is stored in **/etc/bandit_pass/bandit14 and can only be read by user bandit14**. For this level, we don't get the next password, but we get a private SSH key that can be used to log into the next level. Look at the commands that logged we into previous bandit levels, and find out how to use the key for this level.
**Solution:**
Log in using : ssh bandit13@bandit.labs.overthewire.org -p 2220

```
bandit13@bandit:~$ ls
sshkey.private
bandit13@bandit:~$ cat sshkey.private
-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEAxkkOE83W2cOT7IWhFc9aPaaQmQDdgzuXCv+ppZHa++buSkN+
gg0tcr7Fw8NLGa5+Uzec2rEg0WmeevB13AIoYp0MZyETq46t+jk9puNwZwIt9XgB
ZufGtZEwWbFWw/vVLNwOXBe4UWStGRWzgPpEeSv5Tb1VjLZIBdGphTIK22Amz6Zb
ThMsiMnyJafEwJ/T8PQO3myS91vUHEuoOMAzoUID4kN0MEZ3+XahyK0HJVq68KsV
```

```
bandit13@bandit:~$ nano sshkey.private
```

```
┌──(kali㉿kali)-[~]
└─$ chmod 600 sshkey.private

┌──(kali㉿kali)-[~]
└─$ ssh -i sshkey.private bandit14@bandit.labs.overthewire.org -p 2220
```

```
bandit14@bandit:~$ cat /etc/bandit_pass/bandit14
MU4VWeTyJk8ROof1qqmcBPaLh7lDCPvS
```

**Username:** bandit14
**Password:** MU4VWeTyJk8ROof1qqmcBPaLh7lDCPvS

**Bandit Level 14 - Level 15**
**Task:**
The password for the next level can be retrieved by submitting the password of the current level to **port 30000 on localhost**.
**Solution:**
Log in using : ssh bandit14@bandit.labs.overthewire.org -p 2220
**Use nc (netcat) - made exactly for talking to ports.**
1)Sending Bandit14's password to a local service running on port 30000.
2)connect to localhost on port 30000 using: nc localhost 30000

```
bandit14@bandit:~$ nc localhost 30000
MU4VWeTyJk8ROof1qqmcBPaLh7lDCPvS
Correct!
8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo
```

**Username:** bandit15
**Password:** 8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo

**Bandit Level 15 - Level 16**
**Task:**
The password for the next level can be retrieved by submitting the password of the current level to **port 30001 on localhost** using SSL/TLS encryption.
**Helpful note:** Getting "DONE", "RENEGOTIATING" or "KEYUPDATE"? Read the "CONNECTED COMMANDS" section in the manpage.
**Solution:**
Log in using : ssh bandit15@bandit.labs.overthewire.org -p 2220
To send Bandit15's password to port 30001 on localhost,
but this time the connection MUST use SSL/TLS encryption.
**nc won't work**
**We use openssl**

```
bandit15@bandit:~$ cat /etc/bandit_pass/bandit15
8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo
bandit15@bandit:~$ openssl s_client -connect localhost:30001
CONNECTED(00000003)
```

**Paste password of bandit 15**

```
read R BLOCK
8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo
Correct!
kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx

closed
```

**Username:** bandit16
**Password:** kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx

**Bandit Level 16 - Level 17**
**Task:**
The credentials for the next level can be retrieved by submitting the password of the current level to **a port on localhost in the range 31000 to 32000**. First find out which of these ports have a server listening on them. Then find out which of those speak SSL/TLS and which don't. There is only 1 server that will give the next credentials, the others will simply send back to we whatever we send to it.

1. Scan ports **31000–32000** on localhost (nmap -p 31000-32000 localhost)
2. Find **which ports are OPEN**
3. Among those, find **which use SSL/TLS (openssl s_client -connect localhost:31046)**
4. Only **ONE SSL port** will return **next level credentials**
5. Others just **echo back** what we send

**Tools used:**

- nmap - port scanning
- openssl s_client - SSL testing
- ssh - next login

**Solution:**
Log in using : ssh bandit16@bandit.labs.overthewire.org -p 2220

```
bandit16@bandit:~$ cat /etc/bandit_pass/bandit16
kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx
bandit16@bandit:~$ nmap -p 31000-32000 localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-30 18:07 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00024s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE
31046/tcp open  unknown
31518/tcp open  unknown
31691/tcp open  unknown
31790/tcp open  unknown
31960/tcp open  unknown

Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds
bandit16@bandit:~$ openssl s_client -connect localhost:31046
CONNECTED(00000003)
4087F0F7FF7F0000:error:0A0000F4:SSL routines:ossl_statem_client_rea
atem_clnt.c:398:
---
no peer certificate available
---
No client certificate CA names sent
```

```
bandit16@bandit:~$ cat /etc/bandit_pass/bandit16
kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx
bandit16@bandit:~$ openssl s_client -connect localhost:31790 -quiet
Can't use SSL_get_servername
depth=0 CN = SnakeOil
verify error:num=18:self-signed certificate
verify return:1
depth=0 CN = SnakeOil
verify return:1
kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx
Correct!
-----BEGIN RSA PRIVATE KEY-----
MIIEogIBAAKCAQEAvmOkuifmMg6HL2YPIOjon6iWfbp7c3jx34YkYWqUH57SUdyJ
imZzeyGC0gtZPGujUSxiJSWI/oTqexh+cAMTSMlOJf7+BrJObArnxd9Y7YT2bRPQ
```

```
bandit17@bandit:~$ cat /etc/bandit_pass/bandit17
EReVavePLFHtFlFsjn3hyzMlvSuSAcRD
```

**Use ssh private key to log in into bandit 17**
**Username:** bandit17
**Password:** EReVavePLFHtFlFsjn3hyzMlvSuSAcRD

**Bandit Level 17 - Level 18**
**Task:**
There are 2 files in the homedirectory: passwords.old and passwords.new. The password for the next level is in passwords.new and is the only line that has been changed between passwords.old and passwords.new
**Solution:**
Log in using : ssh bandit17@bandit.labs.overthewire.org -p 2220

```
bandit17@bandit:~$ cd ~
bandit17@bandit:~$ ls
passwords.new  passwords.old
bandit17@bandit:~$ diff passwords.old passwords.new
42c42
< BMIOFKM7CRSLI97voLp3TD80NAq5exxk
---
> x2gLTTjFwMOhQ8oWNbMN362QKxfRqGlO
```

**Username:** bandit18
**Password:** x2gLTTjFwMOhQ8oWNbMN362QKxfRqGlO

**Bandit Level 18 - Level 19**
**Task:**
The password for the next level is stored in a file readme in the homedirectory. Unfortunately, someone has modified .bashrc to log we out when we log in with SSH.
**Solution:**
Log in using : ssh bandit18@bandit.labs.overthewire.org -p 2220

```
┌──(kali㉿kali)-[~]
└─$ ssh bandit18@bandit.labs.overthewire.org -p 2220 cat readme

                      _                     _ _ _   
                     | |__   __ _ _ __   __| (_) |_ 
                     | '_ \ / _` | '_ \ / _` | | __|
                     | |_) | (_| | | | | (_| | | |_ 
                     |_.__/ \__,_|_| |_|\__,_|_|\__|


                  This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit18@bandit.labs.overthewire.org's password:
cGWpMaKXVwDUNgPAVJbWYuGHVn9zl3j8
```

**What this does:**
- SSH logs in
- Immediately runs cat readme
- Does not start the interactive shell
- So .bashrc doesn't get a chance to kick we out

**Username:** bandit19
**Password:** cGWpMaKXVwDUNgPAVJbWYuGHVn9zl3j8

**Bandit Level 19 - Level 20**
**Task:**
There is a **setuid binary** in your home directory.When we run it, it runs **as bandit20** (not as bandit19).Using that, we can **read bandit20's password** from: /etc/bandit_pass/bandit20
**Solution:**
Log in using : ssh bandit19@bandit.labs.overthewire.org -p 2220

```
bandit19@bandit:~$ ls -l
total 16
-rwsr-x--- 1 bandit20 bandit19 14884 Oct 14 09:26 bandit20-do
bandit19@bandit:~$ ./bandit20-do
Run a command as another user.
  Example: ./bandit20-do whoami
bandit19@bandit:~$ ./bandit20-do cat /etc/bandit_pass/bandit20
0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbYO
```

Notice the s in permissions: -rwsr-x--- That s = setuid - runs as bandit20
- **setuid** lets a program run with owner's privileges
- Here, the binary is owned by bandit20

- So we temporarily act as bandit20 for that command
- This is how many privilege escalation bugs happen in real systems

**Username:** bandit20
**Password:** 0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbYO

**Bandit Level 20 - Level 21**
**Task:**
There is a **setuid binary** (runs as bandit21).We give it a **port number**.It connects to **localhost:PORT**. It **reads one line** from that connection. If that line == **bandit20 password**.Then it sends back the **bandit21 password**
**Solution:**
Log in using : ssh bandit20@bandit.labs.overthewire.org -p 2220
**nc -lvp 1234**

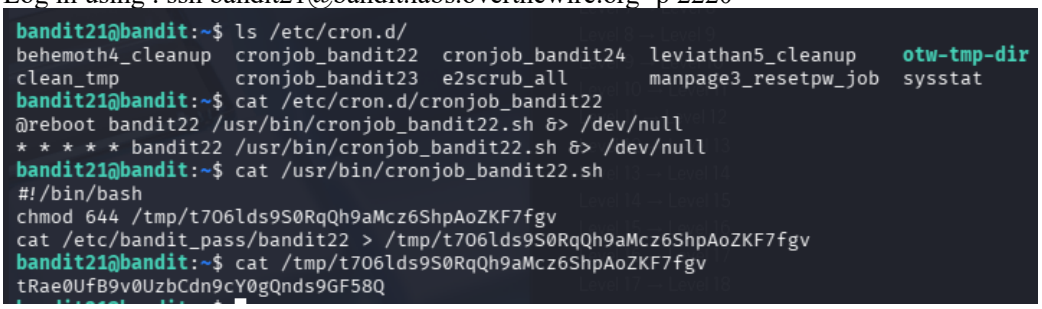
- Starts a listening server on port 1234
- -l = listen
- -v = show connection details
- -p 1234 = use port 1234
- It just waits for a connection

**./suconnect 1234**

- This program connects to your local port 1234
- It talks to your nc listener
- It asks for bandit20 password
- If the password is correct, it sends back bandit21 password

**Username:** bandit21
**Password:** EeoULMCra2q0dSkYj561DX7s1CpBuOBt

**Bandit Level 21 - Level 22**
**Task:**
There's a **cron job** running automatically. Its config is in: /etc/cron.d/. We need to: See which command is being run, Follow it and Find the password for bandit22
A **cron job** is a **task that runs automatically at a scheduled time**.
**Solution:**
Log in using : ssh bandit21@bandit.labs.overthewire.org -p 2220




Found cron job script. Saw it copies password to /tmp/... .We read that file
**Username:** bandit22
**Password:** tRae0UfB9v0UzbCdn9cY0gQnds9GF58Q

**Bandit Level 22 - Level 23**
**Task:**
A program is running automatically at regular intervals from cron, the time-based job scheduler. Look in /etc/cron.d/ for the configuration and see what command is being executed.
**Solution:**
Log in using : ssh bandit22@bandit.labs.overthewire.org -p 2220

```
bandit22@bandit:~$ ls /etc/cron.d/
behemoth4_cleanup  cronjob_bandit22  cronjob_bandit24  leviathan5_cleanup
clean_tmp          cronjob_bandit23  e2scrub_all       manpage3_resetpw_job
bandit22@bandit:~$ cat /etc/cron.d/cronjob_bandit23
@reboot bandit23 /usr/bin/cronjob_bandit23.sh  &> /dev/null
* * * * * bandit23 /usr/bin/cronjob_bandit23.sh  &> /dev/null
bandit22@bandit:~$ cat /usr/bin/cronjob_bandit23.sh
#!/bin/bash

myname=$(whoami)
mytarget=$(echo I am user $myname | md5sum | cut -d ' ' -f 1)

echo "Copying passwordfile /etc/bandit_pass/$myname to /tmp/$mytarget"

cat /etc/bandit_pass/$myname > /tmp/$mytarget
bandit22@bandit:~$ cat /tmp/$mytarget
cat: /tmp/: Permission denied
bandit22@bandit:~$ echo I am user bandit23 | md5sum
8ca319486bfbbc3663ea0fbe81326349  -
bandit22@bandit:~$ cat /tmp/8ca319486bfbbc3663ea0fbe81326349
0Zf11ioIjMVN551jX3CmStKLYqjk54Ga
```

cat script - see what cron does
echo ... | md5sum - get filename
cat /tmp/<hash> - get password
**Username:** bandit23
**Password:** 0Zf11ioIjMVN551jX3CmStKLYqjk54Ga

**Bandit Level 23 - Level 24**
**Task:**
A program is running automatically at regular intervals from **cron**, the time-based job scheduler. Look in **/etc/cron.d/** for the configuration and see what command is being executed.
**Solution:**
Log in using : ssh bandit23@bandit.labs.overthewire.org -p 2220

```
bandit23@bandit:~$ ls /etc/cron.d/
behemoth4_cleanup  cronjob_bandit22  cronjob_bandit24  leviathan5_
clean_tmp          cronjob_bandit23  e2scrub_all       manpage3_re
bandit23@bandit:~$ cat /etc/cron.d/cronjob_bandit24
@reboot bandit24 /usr/bin/cronjob_bandit24.sh &> /dev/null
* * * * * bandit24 /usr/bin/cronjob_bandit24.sh &> /dev/null
bandit23@bandit:~$ cat /usr/bin/cronjob_bandit24.sh
#!/bin/bash
shopt -s nullglob
myname=$(whoami)

cd /var/spool/"$myname"/foo || exit
echo "Executing and deleting all scripts in /var/spool/$myname/foo
for i in * .*;
do
    if [ "$i" != "." ] && [ "$i" != ".." ];
    then
        echo "Handling $i"
        owner="$(stat --format "%U" "./$i")"
        if [ "${owner}" = "bandit23" ] && [ -f "$i" ]; then
            timeout -s 9 60 "./$i"
        fi
        rm -rf "./$i"
    fi
done
bandit23@bandit:~$ /var/spool/bandit24/foo
-bash: /var/spool/bandit24/foo: Is a directory
bandit23@bandit:~$ cd /tmp
bandit23@bandit:/tmp$ nano steal.sh
Unable to create directory /home/bandit23/.local/share/nano/: No s
It is required for saving/loading search history or cursor positio

bandit23@bandit:/tmp$ chmod +x steal.sh
bandit23@bandit:/tmp$ cp steal.sh /var/spool/bandit24/foo/
bandit23@bandit:/tmp$ cat /tmp/b24pass.txt
gb8KRRCsshuZXI0tUuR6ypOFjiZbf3G8
```

ls /etc/cron.d
- To see scheduled cron jobs for Bandit levels.
cat /usr/bin/cronjob_bandit24.sh
- To read what the cron script does and where it runs files from.
(We saw it runs scripts from /var/spool/bandit24/foo and deletes them)
- So we know where to drop your script.
cd /tmp
- To work in a writable directory.
nano steal.sh
- To create a script that copies the password file.
chmod +x steal.sh
- To make the script executable (cron needs this).
cp steal.sh /var/spool/bandit24/foo/
- To place your script where the cron job will execute it.

(Wait ~1 minute)
- Cron runs your script automatically.
cat /tmp/yourfile.txt (or similar)
- To read the copied password for the next level.
**Username:** bandit24
**Password:** gb8KRRCsshuZXI0tUuR6ypOFjiZbf3G8

**Bandit Level 24 - Level 25**
**Task:**
A daemon is listening on port 30002 and will give we the password for bandit25 if given the password for bandit24 and a secret numeric 4-digit pincode. There is no way to retrieve the pincode except by going through all of the 10000 combinations, called brute-forcing.( 0000 to 9999)
**Solution:**
Log in using : ssh bandit24@bandit.labs.overthewire.org -p 2220

```
bandit24@bandit:~$ nc localhost 30002
I am the pincode checker for user bandit25. Please enter the password for user bandit24 and the secret pincode on a s
ingle line, separated by a space.
gb8KRRCsshuZXI0tUuR6ypOFjiZbf3G8 0000
Wrong! Please enter the correct current password and pincode. Try again.
```

Checking the working

```
bandit24@bandit:~$ for i in {0000..9999}; do
  echo "gb8KRRCsshuZXI0tUuR6ypOFjiZbf3G8 $i"
done | nc localhost 30002
-bash: fork: retry: Resource temporarily unavailable
I am the pincode checker for user bandit25. Please enter the password for
ingle line, separated by a space.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
```

Using loop till getting correct password

```
Wrong! Please enter the correct current password and pincode. Try a
Wrong! Please enter the correct current password and pincode. Try a
Wrong! Please enter the correct current password and pincode. Try a
Correct!
The password of user bandit25 is iCi86ttT4KSNe1armKiwbQNmB3YJP3q4
```

**Username:** bandit25
**Password:** iCi86ttT4KSNe1armKiwbQNmB3YJP3q4

**Bandit Level 25 - Level 26**
**Task:**
Logging in to bandit26 from bandit25 should be fairly easy… The shell for user bandit26 is not **/bin/bash**, but something else. Find out what it is, how it works and how to break out of it.
**Solution:**
Log in using : ssh bandit25@bandit.labs.overthewire.org -p 2220
Copy the SSH key from bandit25 to your own machine:  cat bandit26.sshkey
Create the key file on your machine and set permission: chmod 600 bandit26.sshkey
Make terminal window small height (to force pager)
Login using the key: ssh -p 2220 -i bandit26.sshkey bandit26@bandit.labs.overthewire.org
When output opens in pager, press: v

```
:shell
bandit26@bandit:~$ cat /etc/bandit_pass/bandit26
s0773xxkk0MXfdqOfPRVr9L3jJBUOgCZ
bandit26@bandit:~$
```

In vi, set shell and spawn it
- :set shell=/bin/bash
- :shell

Read the password:  cat /etc/bandit_pass/bandit26
**Username:** bandit26
**Password:** s0773xxkk0MXfdqOfPRVr9L3jJBUOgCZ

**Bandit Level 26 - Level 27**
**Task:**
Good job getting a shell! Now hurry and grab the password for bandit27! Bandit27 password is inside a git repository

**Solution:**
Log in using : ssh bandit25@bandit.labs.overthewire.org -p 2220 then v shell

```
bandit26@bandit:~$ ls
bandit27-do  text.txt
bandit26@bandit:~$ file bandit27-do
bandit27-do: setuid ELF 32-bit LSB executable, Intel 80386, vers:
-linux.so.2, BuildID[sha1]=35d353cf6d732f515a73f50ed205265fe1e68
bandit26@bandit:~$ ./bandit27-do
Run a command as another user.
  Example: ./bandit27-do id
bandit26@bandit:~$ ./bandit27-do whoami
bandit27
bandit26@bandit:~$ cat /etc/bandit_pass/bandit27
cat: /etc/bandit_pass/bandit27: Permission denied
bandit26@bandit:~$ ./bandit27-do cat /etc/bandit_pass/bandit27
upsNCc7vzaRDx6oZC6GiR6ERwe1MowGB
```

**Username:** bandit27
**Password:** upsNCc7vzaRDx6oZC6GiR6ERwe1MowGB

**Bandit Level 27 - Level 28**
**Task:**
There is a git repository at ssh://bandit27-git@bandit.labs.overthewire.org/home/bandit27-git/repo via the port 2220. The password for the user bandit27-git is the same as for the user bandit27.
**From your local machine (not the OverTheWire machine!), clone the repository and find the password for the next level. This needs git installed locally on your machine**
**Solution:**
Log in using : ssh bandit27@bandit.labs.overthewire.org -p 2220

```
bandit27@bandit:~$ mkdir /tmp/b27 && cd /tmp/b27
bandit27@bandit:/tmp/b27$ exit
logout
Connection to bandit.labs.overthewire.org closed.

┌──(kali㉿kali)-[~]
└─$ git clone ssh://bandit27-git@bandit.labs.overthewire.org:2220/home/bandit27-git/repo
Cloning into 'repo'...
                         _
                        | |__   __ _ _ __   __ _| (_) ___
                        | '_ \ / _` | '_ \ / _` | | |/ __|
                        | |_) | (_| | | | | (_| | | | (__
                        |_.__/ \__,_|_| |_|\__,_|_|_|\___|


                       This is an OverTheWire game server.
             More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit27-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
Receiving objects: 100% (3/3), done.

┌──(kali㉿kali)-[~]
└─$ cd repo

┌──(kali㉿kali)-[~/repo]
└─$ ls
README

┌──(kali㉿kali)-[~/repo]
└─$ cat README
The password to the next level is: Yz9IpL0sBcCeuG7m9uQFt8ZNpS4HZRcN
```

mkdir /tmp/b27 && cd /tmp/b27 - create & enter folder
git clone ssh://bandit27-git@bandit.labs.overthewire.org:2220/home/bandit27-git/repo - clone repo (use port 2220)
Enter bandit27 password when asked
**Username:** bandit28
**Password:** Yz9IpL0sBcCeuG7m9uQFt8ZNpS4HZRcN

**Bandit Level 28 - Level 29**
**Task:**
There is a git repository at ssh://bandit28-git@bandit.labs.overthewire.org/home/bandit28-git/repo via the port 2220. The password for the user bandit28-git is the same as for the user bandit28.
**Solution:**
Log in using : ssh bandit28@bandit.labs.overthewire.org -p 2220
(Open your own Kali)
mkdir /tmp/b28 && cd /tmp/b28 - create & enter folder
git clone ssh://bandit28-git@bandit.labs.overthewire.org:2220/home/bandit28-git/repo - clone repo
Enter bandit28 password when asked
cd repo - enter repo
ls - list files

cat README - see hint (password is NOT here)
git log - check commit history
git show <commit_id> - view older commit
Find password for bandit29 in previous commit

```
┌──(kali㉿kali)-[~]
└─$ mkdir /tmp/b28 && cd /tmp/b28

┌──(kali㉿kali)-[/tmp/b28]
└─$ git clone ssh://bandit28-git@bandit.labs.overthewire.org:2220/home/bandit28-git/repo
Cloning into 'repo'...

                         _                 _   _ _   _
                        | |__   __ _ _ __ __| (_) | |_
                        | '_ \ / _` | '_ \ / _` | | | __|
                        | |_) | (_| | | | | (_| | | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit28-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 9 (delta 2), reused 0 (delta 0), pack-reused 0
Receiving objects: 100% (9/9), done.
Resolving deltas: 100% (2/2), done.

┌──(kali㉿kali)-[/tmp/b28]
└─$ cd repo

┌──(kali㉿kali)-[/tmp/b28/repo]
└─$ ls
README.md

┌──(kali㉿kali)-[/tmp/b28/repo]
└─$ cat README.md
# Bandit Notes
Some notes for level29 of bandit.

## credentials

- username: bandit29
- password: xxxxxxxxxx
```

```
┌──(kali㉿kali)-[/tmp/b28/repo]
└─$ git log
commit b5ed4b5a3499533c2611217c8780e8ead48609f6 (HEAD -> master, origin/master, origin/HEAD)
Author: Morla Porla <morla@overthewire.org>
Date:   Tue Oct 14 09:26:24 2025 +0000

    fix info leak

commit 8b7c651b37ce7a94633b7b7b7c980ded19a16e4f
Author: Morla Porla <morla@overthewire.org>
Date:   Tue Oct 14 09:26:24 2025 +0000

    add missing data

commit 6d8e5e607602b597ade7504a550a29ba03f2cca0
Author: Ben Dover <noone@overthewire.org>
Date:   Tue Oct 14 09:26:24 2025 +0000

    initial commit of README.md

┌──(kali㉿kali)-[/tmp/b28/repo]
└─$ git show 8b7c651b37ce7a94633b7b7b7c980ded19a16e4f
commit 8b7c651b37ce7a94633b7b7b7c980ded19a16e4f
Author: Morla Porla <morla@overthewire.org>
Date:   Tue Oct 14 09:26:24 2025 +0000

    add missing data

diff --git a/README.md b/README.md
index 7ba2d2f..d4e3b74 100644
--- a/README.md
+++ b/README.md
@@ -4,5 +4,5 @@ Some notes for level29 of bandit.
 ## credentials

 - username: bandit29
-- password: <TBD>
+- password: 4pT1t5DENaYuqnqvadYs1oE4QLCdjmJ7
```

**Username:** bandit29
**Password:** 4pT1t5DENaYuqnqvadYs1oE4QLCdjmJ7

**Bandit Level 29 - Level 30**
**Task:**
There is a git repository at ssh://bandit29-git@bandit.labs.overthewire.org/home/bandit29-git/repo via the port 2220. The password for the user bandit29-git is the same as for the user bandit29.
**Solution:**
Log in using : ssh bandit29@bandit.labs.overthewire.org -p 2220 (Open your own Kali)
mkdir /tmp/b29 && cd /tmp/b29 - create & enter folder
git clone ssh://bandit29-git@bandit.labs.overthewire.org:2220/home/bandit29-git/repo - clone repo
Enter bandit29 password when asked

```
┌──(kali㉿kali)-[/tmp/b29]
└─$ cd repo

┌──(kali㉿kali)-[/tmp/b29/repo]
└─$ ls
README.md

┌──(kali㉿kali)-[/tmp/b29/repo]
└─$ cat README.md
# Bandit Notes
Some notes for bandit30 of bandit.

## credentials

- username: bandit30
- password: <no passwords in production!>

┌──(kali㉿kali)-[/tmp/b29/repo]
└─$ git branch -a
* master
  remotes/origin/HEAD -> origin/master
  remotes/origin/dev
  remotes/origin/master
  remotes/origin/sploits-dev

┌──(kali㉿kali)-[/tmp/b29/repo]
└─$ git checkout dev
branch 'dev' set up to track 'origin/dev'.
Switched to a new branch 'dev'

┌──(kali㉿kali)-[/tmp/b29/repo]
└─$ ls
code  README.md

┌──(kali㉿kali)-[/tmp/b29/repo]
└─$ cat README.md
# Bandit Notes
Some notes for bandit30 of bandit.

## credentials

- username: bandit30
- password: qp30ex3VLz5MDG1n91YowTv4Q8l7CDZL
```

cd repo - enter repo
ls - list files, cat README - see it says “no password here”
git branch -a - list all branches
git checkout dev - switch to dev branch
ls - list files, cat README - get bandit30 password
**Username:** bandit30
**Password:** qp30ex3VLz5MDG1n91YowTv4Q8l7CDZL

**Bandit Level 30 - Level 31**
**Task:**
There is a git repository at ssh://bandit30-git@bandit.labs.overthewire.org/home/bandit30-git/repo via the port 2220. The password for the user bandit30-git is the same as for the user bandit30.
**Solution:**
Log in using : ssh bandit30@bandit.labs.overthewire.org -p 2220 **(Open your own Kali)**

```
┌──(kali㉿kali)-[~]
└─$ mkdir /tmp/b30 && cd /tmp/b30

┌──(kali㉿kali)-[/tmp/b30]
└─$ git clone ssh://bandit30-git@bandit.labs.overthewire.org:2220/home/bandit30-git/repo
Cloning into 'repo' ...
                         _                     _ _ _   
                        | |__   __ _ _ __   __| (_) |_ 
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_ 
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit30-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
Receiving objects: 100% (4/4), done.

┌──(kali㉿kali)-[/tmp/b30]
└─$ cd repo

┌──(kali㉿kali)-[/tmp/b30/repo]
└─$ ls
README.md

┌──(kali㉿kali)-[/tmp/b30/repo]
└─$ cat README.md
just an epmty file ... muahaha

┌──(kali㉿kali)-[/tmp/b30/repo]
└─$ git tag
secret

┌──(kali㉿kali)-[/tmp/b30/repo]
└─$ git show secret
fb5S2xb7bRyFmAvQYQGEqsbhVyJqhnDy
```

mkdir /tmp/b30 && cd /tmp/b30 - create & enter folder
git clone ssh://bandit30-git@bandit.labs.overthewire.org:2220/home/bandit30-git/repo - clone repo
Enter bandit30 password when asked
cd repo - enter repo, ls - list files
cat README - see message (no password shown)
git tag - list tags, git show secret - view tag named secret
Copy the bandit31 password from output
**Username:** bandit31
**Password:** fb5S2xb7bRyFmAvQYQGEqsbhVyJqhnDy

**Bandit Level 31 - Level 32**
**Task:**
There is a git repository at ssh://bandit31-git@bandit.labs.overthewire.org/home/bandit31-git/repo via the port 2220. The password for the user bandit31-git is the same as for the user bandit31.
**Solution:**
Log in using : ssh bandit31@bandit.labs.overthewire.org -p 2220
git clone ... - cloned the repo, cd repo - entered repo

```
┌──(kali㉿kali)-[/tmp/b31/repo]
└─$ cat README.md
This time your task is to push a file to the remote repository.

Details:
    File name: key.txt
    Content: 'May I come in?'
    Branch: master


┌──(kali㉿kali)-[/tmp/b31/repo]
└─$ cat .gitignore
*.txt

┌──(kali㉿kali)-[/tmp/b31/repo]
└─$ echo "May I come in?" > key.txt

┌──(kali㉿kali)-[/tmp/b31/repo]
└─$ git add -f key.txt

┌──(kali㉿kali)-[/tmp/b31/repo]
└─$ git commit -m "add key"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'kali@kali.(none)')
```

cat README.md - read instructions, cat .gitignore - saw *.txt is ignored
echo "May I come in?" > key.txt - created required file
git add -f key.txt - force-added ignored file, git commit -m "add key" - failed (no name/email set)
Git didn't commit - nothing got pushed
git config --global user.name "kali" - set git name
git config --global user.email "kali@kali" - set git email
git commit -m "add key" - commit worked
git push - pushed to server
Server printed bandit32 password (even though push was rejected)

```
┌──(kali㉿kali)-[/tmp/b31/repo]
└─$ git config --global user.name "kali"

┌──(kali㉿kali)-[/tmp/b31/repo]
└─$ git config --global user.email "kali@kali"

┌──(kali㉿kali)-[/tmp/b31/repo]
└─$ git commit -m "add key"
[master e0c3eb4] add key
 1 file changed, 1 insertion(+)
 create mode 100644 key.txt

┌──(kali㉿kali)-[/tmp/b31/repo]
└─$ git push
```

**Username:** bandit32
**Password:** 3O9RfhqyAlVBEZpVb6LYStshZoqoSx5K

**Bandit Level 32 - Level 33**
**Task:**
After all this git stuff, it's time for another escape. Good luck!
**Solution:**
Log in using : ssh bandit32@bandit.labs.overthewire.org -p 2220

```
WELCOME TO THE UPPERCASE SHELL
>> $0
$
$ cat /etc/bandit_pass/bandit33
tQdtbs5D5i2vJwkO8mEyYEyTL8izoeJ0
```

We get **UPPERCASE shell** (it breaks normal commands) - It converts everything to uppercase
Type: $0 = name of current shell - launches a normal shell - escapes the restricted shell
Now run: cat /etc/bandit_pass/bandit33
Reads the file that stores bandit33 password
**Username:** bandit33
**Password:** tQdtbs5D5i2vJwkO8mEyYEyTL8izoeJ0